E-JUST

الجامعة المصرية اليابانية للعلوم و التكنولوجيا
エジプト日本科学技術大学
EGYPT-JAPAN UNIVERSITY OF SCIENCE AND TECHNOLOGY

School of Electronics, Communications, and Computer Engineering

Department of Electronics, and Communications Engineering

ECE 212: ELECTRONICS ENGINEERING LAB

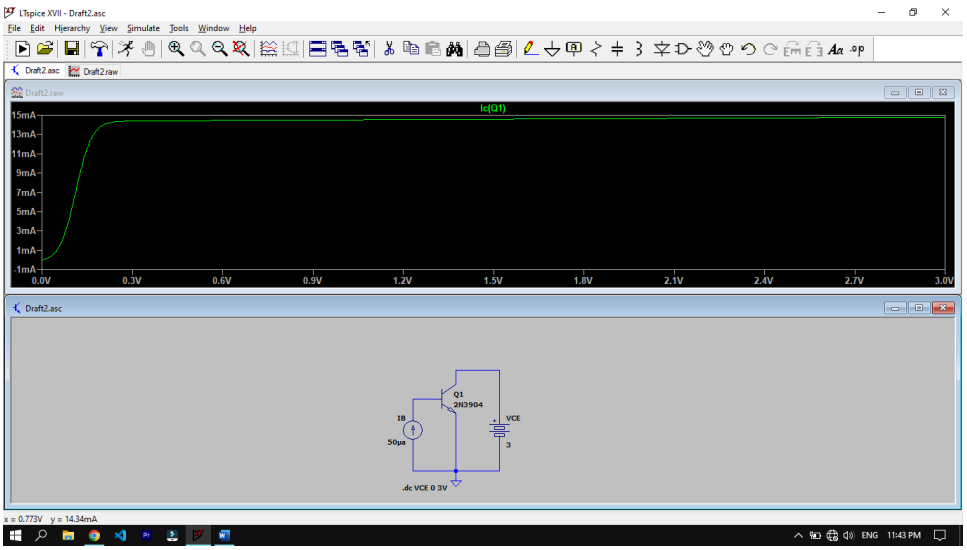
Experiment 04 - Bipolar Junction Transistor Characteristics

Student Name: Abdelrahman Hisham Abdelrahman
Student ID: ……..120210254……
Section: ……12…….

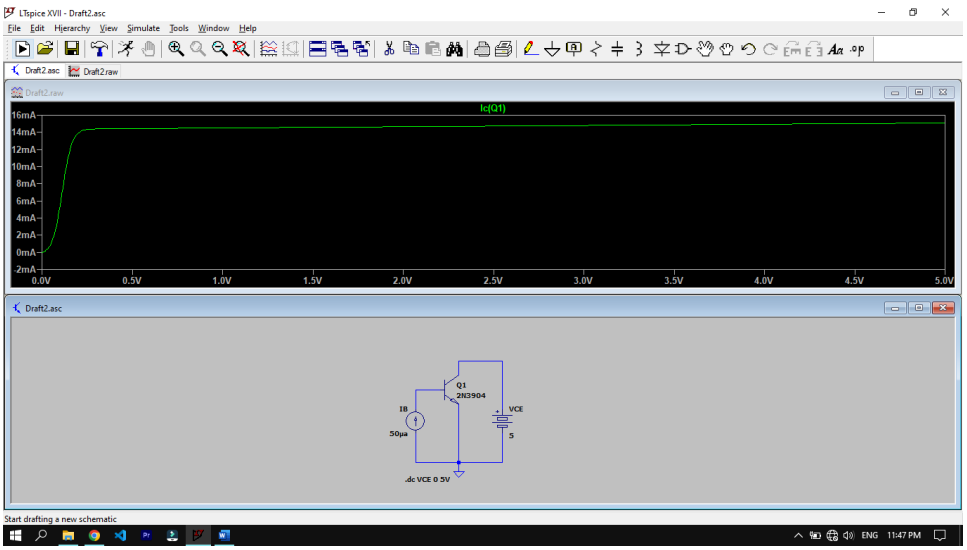
TO:

Dr. Ahmed H. Abd El-Malek

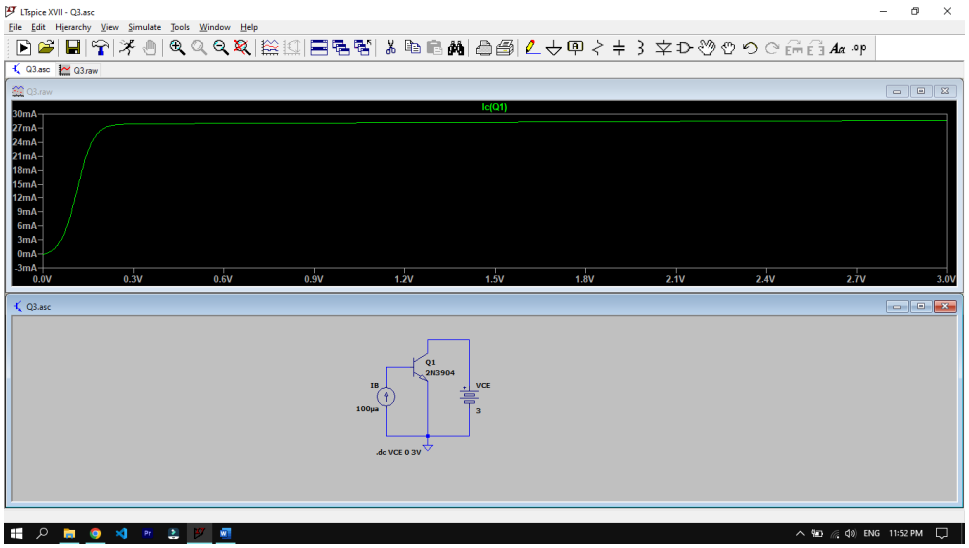
Fall 2022



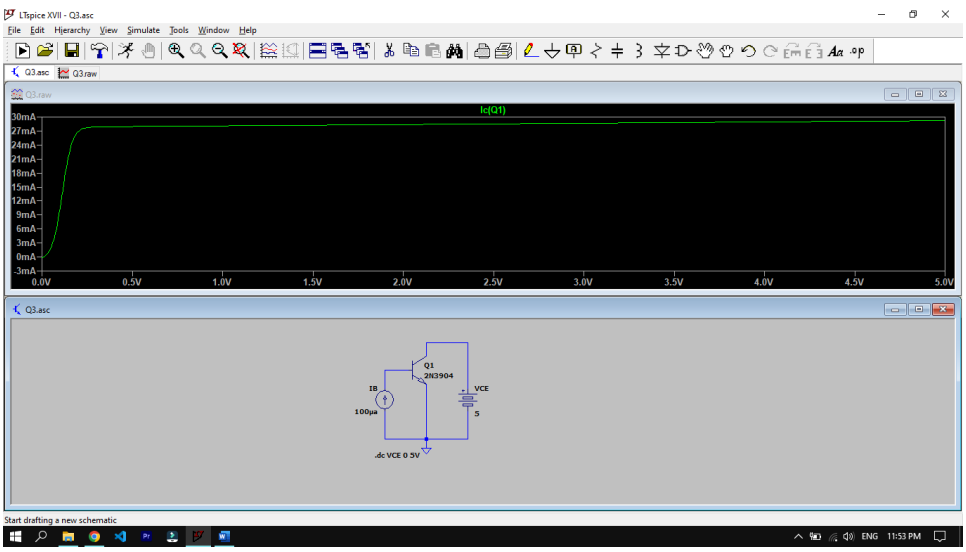
IC=14.8µa
IB=50µa
B=IC/IB=296
α=β/β+1=0.9966



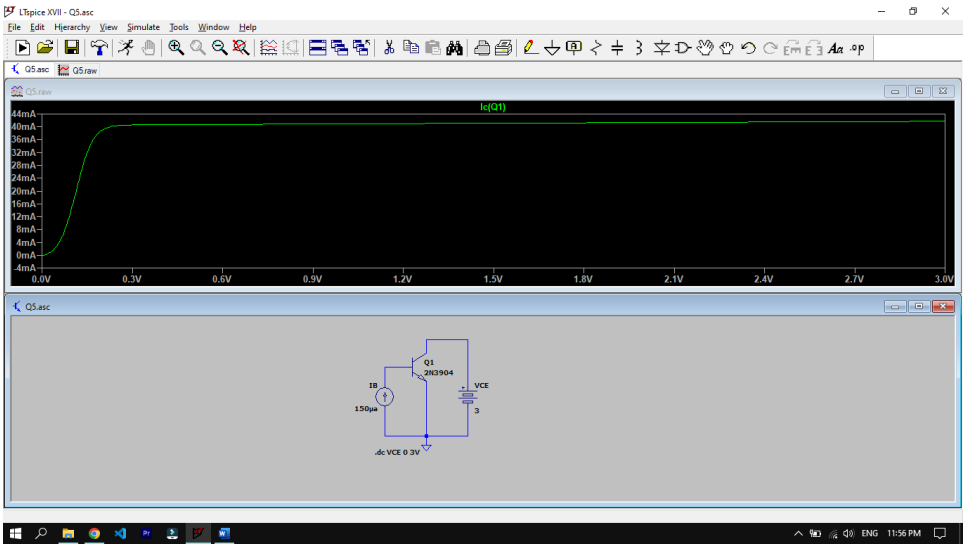
IC=15µa
IB=50µa
B=IC/IB=300
α=β/β+1=0.99667



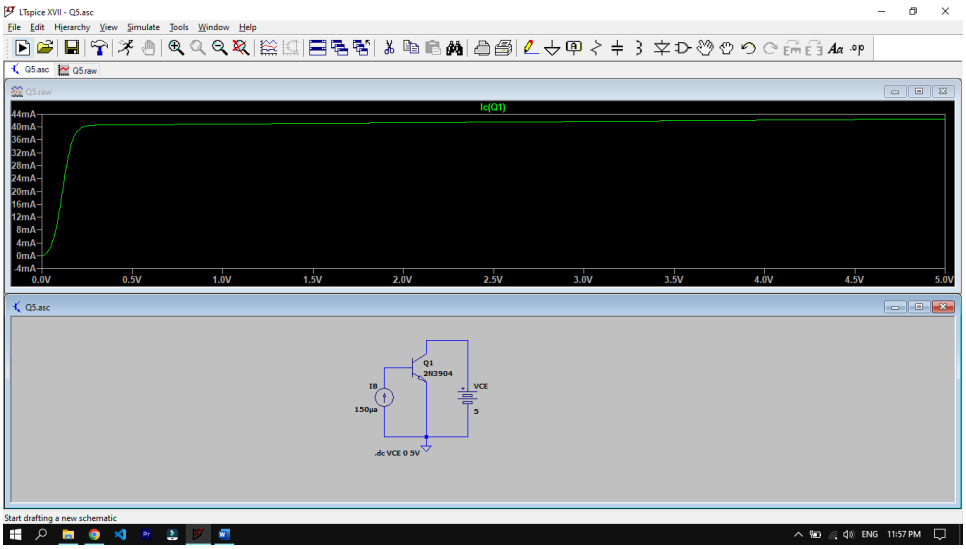
IC=28.65µa
IB=100µa
B=IC/IB=286.5
α=β/β+1=0.99652



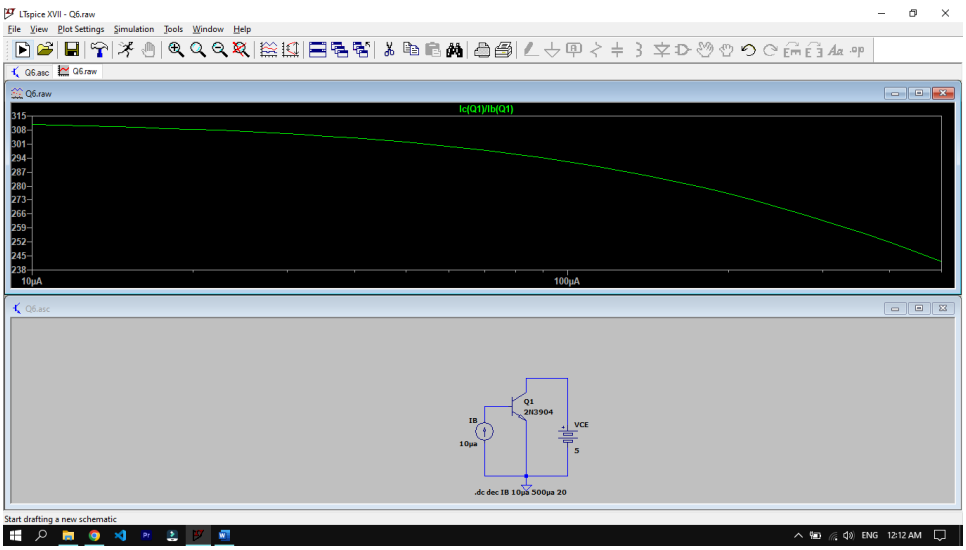
IC=28.652µa
IB=100µa
B=IC/IB=286.52
α=β/β+1=0.996521



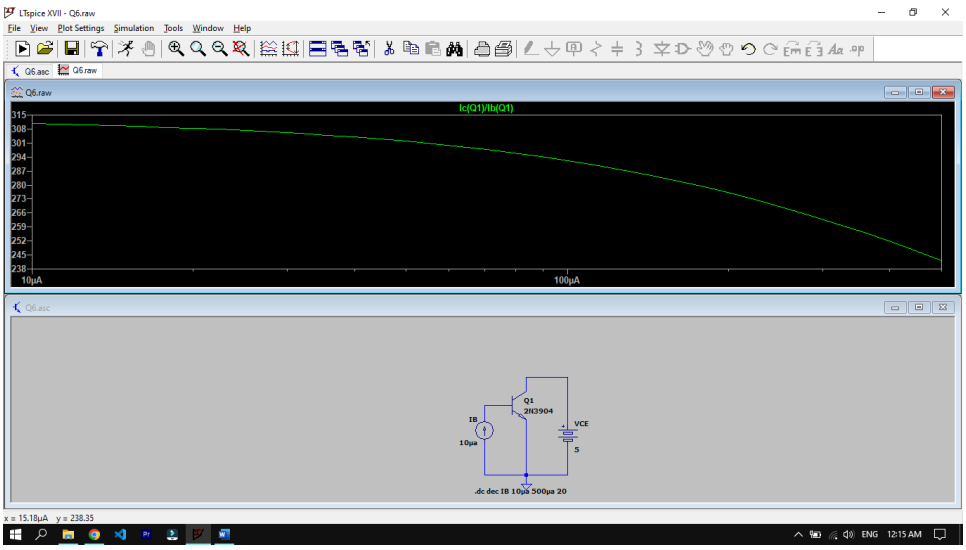
IC=40.68µa
IB=150µa
B=IC/IB=271.2
α=β/β+1=0.99632



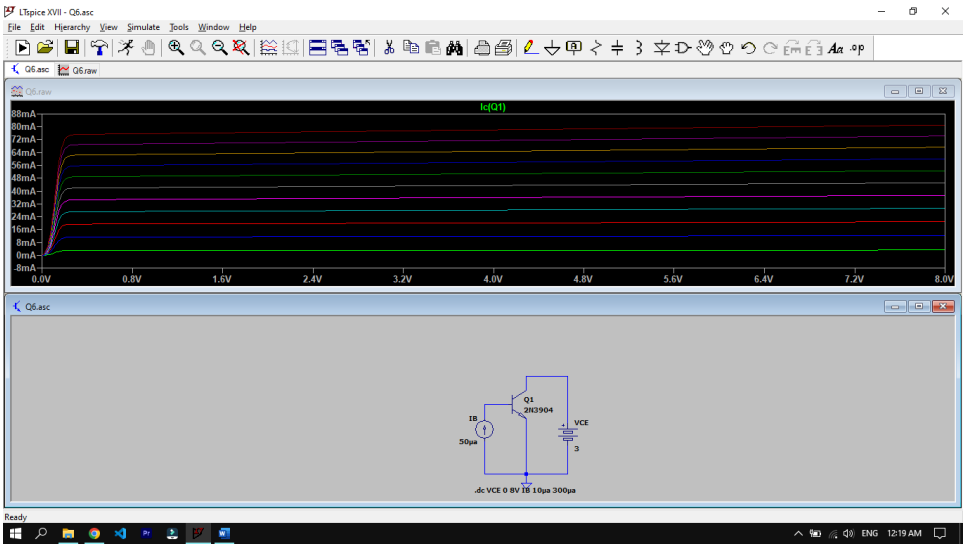
IC=41.3µa
IB=150µa
B=IC/IB=275.33
α=β/β+1=0.9963811



WITH _IC_IN _X_AXIS
AT IC = 3.10635µa
MAX[B]=310.635
α=0.996791



WITH _IB_IN _X_AXIS
AT IB= 10.02µa :
IC =3.106µa
MAX[B]=310.6
α=0.99679



IC_VC CHARACTERISTICS